



ET502 Measurement and Control (M1 + LM1)

TERM: Fall 2025 (8/25/25 – 12/8/25)

CREDITS: 4 Credits

PREREQUISITES: MATH 418

DEPARTMENT/PROGRAM:

Department of Applied Engineering and Sciences / Engineering Technology

COURSE DESCRIPTION:

The intention of this course is to develop skills in the areas of:

- Data Collection
- Data Graphing
- Fundamentals of Control systems
- Measurement
- Instrumentation
- Statistical analysis
- Error Estimation

These concepts will be explored in a series of weekly discussions and hands on activities

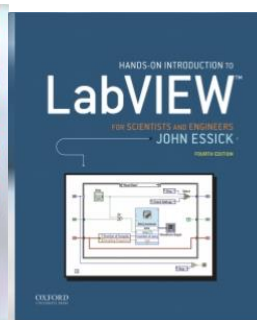
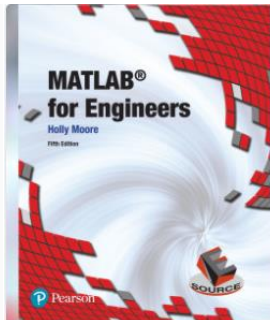
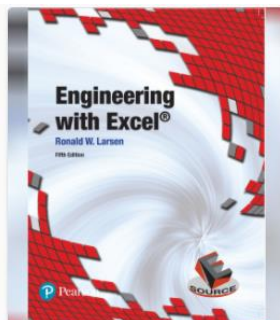
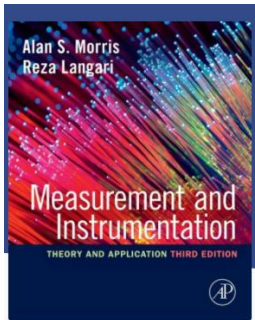
CLASS TIME AND PLACE:

Section	Day	Start Time	End Time	Room
Lecture	Friday	9:10 AM	10:30 AM	P102
Laboratory	Friday	10:40 AM	12:30 PM	P114

REFERENCE BOOKS:

No textbooks will be used, but here are reference books; you are not required to purchase these books.

1. Measurement and Instrumentation, Theory and Application third edition Alan S. Morris; Reza Langari
2. Engineering with Excel, fifth edition Ronald W. Larsen
3. MATLAB for Engineers, fifth edition Holly Moore
4. LabVIEW for Scientists and Engineers, Fourth Edition, John Essick



Required Software Available in Lab P114

Microsoft Excel <https://www.microsoft.com/en-us/microsoft-365/excel>
MATLAB <https://www.mathworks.com/products/matlab.html>
LabView <https://www.ni.com/>

Available for Free Download

<https://td.unh.edu/TDClient/60/Portal/KB/ArticleDet?ID=770>

INSTRUCTOR: Takahide Ohkami, Ph.D. (Lecturer)
Office: P119
Email: takahide.ohkami@unh.edu

OFFICE HOURS: Arranged by appointment.

GRADING:

The grade for this class will be determined by the 13 laboratory activity reports. The grading rubric of each assignment will be described in each assignment.

The grading weights are as follows:

Excel Projects (W1, W2)	17%
MATLAB Projects (W3, W4)	17%
Instrument Projects (W5, W6, W7)	26%
JPL Mars Landing Project (W8)	8%
Ohm's Law Projects (W9, W10)	17%
LabVIEW Projects (W11, W12, W13)	15%

The final grade mode is letter grading:



A: $94 \leq \text{Score}$	A-: $90 \leq \text{Score} < 94$	
B+: $87 \leq \text{Score} < 90$	B: $84 \leq \text{Score} < 87$	B-: $80 \leq \text{Score} < 84$
C+: $77 \leq \text{Score} < 80$	C: $74 \leq \text{Score} < 77$	C-: $70 \leq \text{Score} < 74$
D+: $67 \leq \text{Score} < 70$	D: $64 \leq \text{Score} < 67$	D-: $60 \leq \text{Score} < 64$
F: $\text{Score} < 60$		

If you have questions during the semester, please feel free to ask the instructor. If you are having trouble understanding the material or with the course, please let the instructor know as early as possible.

LABORATORY ASSIGNMENTS:

Laboratory assignments will be assigned throughout the semester. All assignment information will be posted on Canvas with applicable due dates. All assignments must be submitted by the due date. Late assignments will be accepted up to “ONE-WEEK” beyond the announced due date. The score of the late assignments will be reduced by 10% of the full score. Late assignments will not be accepted after the “ONE-WEEK” period and a score of “zero” will be recorded for that assignment.

LABORATORY ASSIGNMENT SUBMISSION REQUIREMENTS:

General requirements for assignments are as follows:

1. Always show all your work. (No partial credit can be given unless steps are clearly shown.)
2. Clearly identify final answers.
3. Work should be legible and performed in a clear organized manner.
4. Turn in lab assignments in electronic form. Electronic form should consist of a single coherent legible document for each single assignment. Do not submit several pieces of electronic items such as phone snapshots for a given single assignment. Scanning may be done using the ET Commons copier/printer.
5. Include a cover page with assignment information on each assignment submitted. This includes the assignment number and type/title of assignment. Be sure to include your name on cover page.
6. Some assignment formats may vary per activity/exercise.

These items facilitate fair grading of the assignment and granting of partial credit where warranted. Working on activity assignments in groups is permitted but each person should submit his/her own individual assignments.

MIDTERM AND FINAL EXAMINATIONS:

There will be no midterm and final examinations.

HOMEWORK ASSIGNMENTS:



There will be no homework assignments.

ATTENDANCE:

The University absent policy is found in the handbook at this website address:

<https://www.unh.edu/student-life/student-rights-rules-responsibilities> Students are responsible for all material covered in class and posted on Canvas. Regular class attendance and participation in discussions is strongly encouraged and expected.

ACADEMIC HONESTY:

In the preparation and presentation of any assigned work – including examinations, tests, quizzes, term papers, reports, papers and other written or oral exercises – every student shall conform to a strict standard of academic honesty. Any attempt to deceive a faculty member or to help another student to do so will be considered a violation of this standard. In all assignments, students must acknowledge the words and/or ideas of others taken from print or electronic media, whether a direct quotation or a paraphrase; any omission of this is dishonest. Cheating on examinations or tests consists of knowingly giving, receiving, or using – or attempting to give, receive, or use – unauthorized assistance during an examination or test. A faculty member may record a grade of “zero” for any assignment on which a student has plagiarized or cheated. For repeat offenses within a single course, the faculty member may record a grade of “F” for the course. Violations of this policy in multiple courses may result in dismissal from the college. See [University Academic Honesty Policy](#) and [Tutorial on Plagiarism](#).

PROFESSIONALISM:

Honesty is expected of all students. Students are expected to complete their own work at all times. When working in an instructor approved group, in specified lab exercises, group members may work together to assist one another when appropriate, but each student is expected to submit their own original work. Plagiarism or cheating is a serious offense and will be treated according to the guidelines outlined in the NHTI Student Handbook.

Students are expected to show proper behavior and respect in the classroom and laboratory. Each student is expected to act as a professional as would be expected in industry. This includes the use of proper language and attitude toward the instructor and other students. Students should be on-time, not exit and re-enter class or lab excessively, and not leave early. Cell phones, text messaging devices, iPods **should be turned off and put away** during lecture and laboratory sessions. Laptops are permitted for **note taking only**, NOT for accessing the internet or email services. Courtesy and respect is expected from all students. **All safety practices and policies must be adhered to at all times, especially in the lab.** Improper behavior or continued violations of any of the above rules will not be tolerated. Violation of the above rules, especially safety rules, will result in suspension from the lab and/or course with appropriate corresponding grade consequences.



STUDENTS WITH DISABILITIES:

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate his/her disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Janessa Zurek, Student Accessibility Coordinator, in the Student Services Suite at 603.641.4383 or jenessa.zurek@unh.edu. Accommodation letters are created by our Coordinator with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of identified accommodations. Faculty have no obligation to provide retroactive accommodations.

UNH STATEMENT ON EMOTIONAL OR MENTAL DISTRESS:

In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers consultation visits in the Academic Advising department (located in the Student Services suites, 4th floor).

Services include:

- Free confidential screening & consultation with a licensed mental health therapist.
- Referrals to mental health or substance misuse treatment.
- And assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

If you would like to connect to counseling services directly, you may do so by contacting The Greater Manchester Mental Health Center at (603) 668 - 4111. If you or someone you know is experiencing a mental health or substance misuse crisis, please contact the Mobile Crisis Response Team (MCRT) at (800) 688 - 3544.

The National Suicide Prevention Lifeline provides 24/7, free and confidential support via phone or chat for people in distress, resources for you or your loved ones, and best practices for professionals. Call (800) 273-TALK (8255).



COURSE SPECIFIC OBJECTIVES OR OUTCOMES:

Students will

1. Learn data collection from various measurements;
2. Learn graphing collected data using Excel and MATLAB;
3. Learn various measurement and control methods;
4. Learn various instrumentation systems used in industries
5. Learn statistical analysis of data collected from measurements; and
6. Learn error estimations and calibration methods.

ABET/ETAC (Accreditation Board for Engineering and Technology/Engineering Technology Accreditation Commission)

I. General Criteria (2019)

Criterion 3. Student Outcomes

The program must have documented student outcomes with a documented and effective process for the periodic review and revision of these student outcomes. For purposes of this section, well-defined activities or problems are practical, narrow in scope, use conventional processes and materials in traditional ways, and require knowledge of standard operating processes. Broadly-defined activities or problems are practical, broad in scope, relatively complex, and involve a variety of resources; use new processes, materials, or techniques in innovative ways; and may require extension of standard operating procedures.

For baccalaureate degree programs, these student outcomes must include, but are not limited to, the following:

- (1) an ability to apply knowledge, techniques, skills and modern tools of mathematics, science, engineering, and technology to solve broadly-defined engineering problems appropriate to the discipline;
- (2) an ability to design systems, components, or processes meeting specified needs for broadly-defined engineering problems appropriate to the discipline;
- (3) an ability to apply written, oral, and graphical communication in broadly-defined technical and non-technical environments; and an ability to identify and use appropriate technical literature;
- (4) an ability to conduct standard tests, measurements, and experiments and to analyze and interpret the results to improve processes; and
- (5) an ability to function effectively as a member as well as a leader on technical teams.

II. PROGRAM CRITERIA



MECHANICAL ENGINEERING TECHNOLOGY

PROGRAM CRITERIA FOR BACCALAUREATE LEVEL PROGRAMS

The curriculum must prepare baccalaureate degree graduates with the knowledge, techniques, skills, and use of modern equipment in mechanical engineering technology. Baccalaureate degree graduates build on the strengths of associate degree programs by gaining proficiency in the analysis, applied design, development, implementation or oversight of more advanced mechanical components, systems or processes. Programs may focus on preparing graduates with in-depth but narrow expertise, while other programs may choose to prepare graduates with a broad spectrum of expertise. The depth and breadth of expertise demonstrated by baccalaureate graduates must support the program educational objectives. The following curricular topics are required (unless the program's faculty and primary constituents approve the substitution of other specific, mechanically-related technical subjects supporting attainment of program educational objectives):

- a. Application of principles of geometric dimensioning and tolerancing;
- b. Use of computer aided drafting and design software;
- c. Perform selection, set-up, and calibration of measurement tools/instrumentation;
- d. Elements of differential and integral calculus;
- e. Manufacturing processes;
- f. Material science and selection;
- g. Solid mechanics (such as statics, dynamics, strength of materials, etc.);
- h. Mechanical system design;
- i. Thermal sciences (such as thermodynamics, fluid mechanics, heat transfer, etc.);
- j. Electrical circuits (ac and dc) and electronic controls;
- k. Application of industry codes, specifications and standards; and
- l. Technical communications typically used in preparation of engineering proposals, reports, and specifications.

ET502: MECHANICAL ENGINEERING TECHNOLOGY PROGRAM OUTCOMES:

This course covers general topics which lead directly to achieve the ABET MET Program Specific Curriculum requirements in Categories "c" (selection, setup of tools and instrumentation), and "j" (electrical and electronics). These outcomes are assessment tools in the General Criteria 4 Continuous Improvement Plan.

Can also be used for Assessment of Student Outcomes (1) and/or (4).